Lab 9 Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

Objectives:

- Configure a wireless router to establish a functioning Wi-Fi network in Cisco Packet Tracer.
- Connect multiple wireless devices to the network and dynamically allocate IP addresses via DHCP.
- Apply WPA2 encryption to maintain secure wireless communication.
- Examine and validate wireless packet flow using Packet Tracer's simulation tools.

Theory:

Basics of Wireless Communication

Wireless networks enable devices to exchange data without the need for physical cables, utilizing radio frequency (RF) signals. A central wireless router serves as the hub, connecting laptops, smartphones, and other wireless clients. The IEEE 802.11 family of standards governs Wi-Fi communication, ensuring compatibility across devices from different manufacturers.

Functions of a Wireless Router

A wireless router typically performs three key roles:

- Access Point (AP): Facilitates wireless connectivity for client devices.
- Routing Device: Manages traffic between the local network and external networks.
- DHCP Server: Automatically assigns IP addresses, simplifying network configuration.

In Cisco Packet Tracer, the WRT300N router model is widely used to simulate wireless networking environments.

Wireless Security – WPA2 Protocol

Unprotected wireless networks are susceptible to unauthorized access. WPA2 (Wi-Fi Protected Access 2) is a widely adopted security standard that employs AES (Advanced Encryption Standard) to provide robust encryption. Enabling WPA2 ensures:

- Verification of authorized users.
- Confidential handling of transmitted information.
- Defense against threats like eavesdropping and intrusions.

Packet Inspection in Cisco Packet Tracer

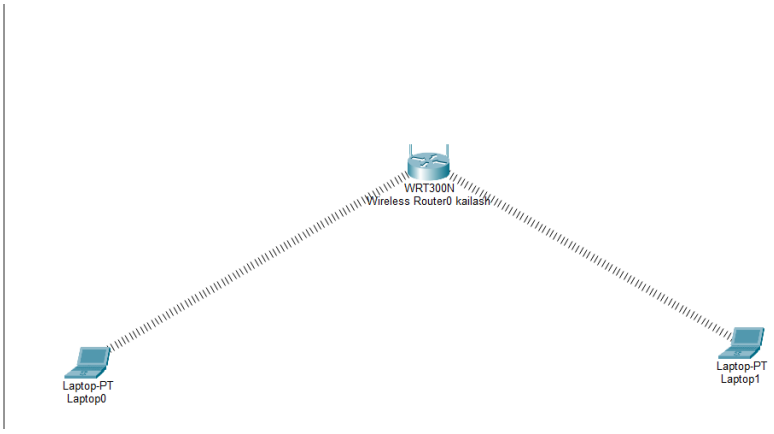
Simulation mode in Packet Tracer allows detailed observation of data movement across the network. Packet analysis can reveal:

- DHCP operations, including IP request and assignment exchanges.
- ARP (Address Resolution Protocol) interactions for device identification.
- Secure transmission of data when WPA2 is implemented.

This approach reinforces theoretical knowledge through hands-on experimentation, allowing users to monitor and understand network behaviors.

NETWORK TOPOLOGY

The lab setup consists of a wireless router connected to multiple wireless clients, including laptops and smartphones. All devices receive IP addresses dynamically from the router's DHCP server. WPA2 security is configured to safeguard the network. Packet Tracer's simulation mode is employed to track packet flow, analyze DHCP and ARP processes, and verify encrypted communication.



Configuration

Device	Ipv4 address	Subnet mask	Default gateway
Labtop 0	192.168.0.103	255.255.255.0	192.168.0.1
Labtop 1	192.168.0.100	255.255.255.0	192.168.0.1

Procedure :

Wireless Router Configuration :

Step 1 : Basic Setup

1. Click on the Wireless Router.

2. Go to GUI → Setup.

3. Configure the following:

• IP Address: 192.168.0.1

• Subnet Mask: 255.255.255.0

• DHCP Server: Enabled

Step 2 : Wireless Configuration

1. Go to Wireless → Basic Wireless Settings.

2. Configure the following parameters:

3. Network Mode: Mixed

• SSID: HCOE

• Channel: Auto

4. Click Save Settings.

Step 3 : Wireless Security

1. Go to Wireless → Wireless Security.

2. Configure the following:

• Security Mode: WPA2-Personal

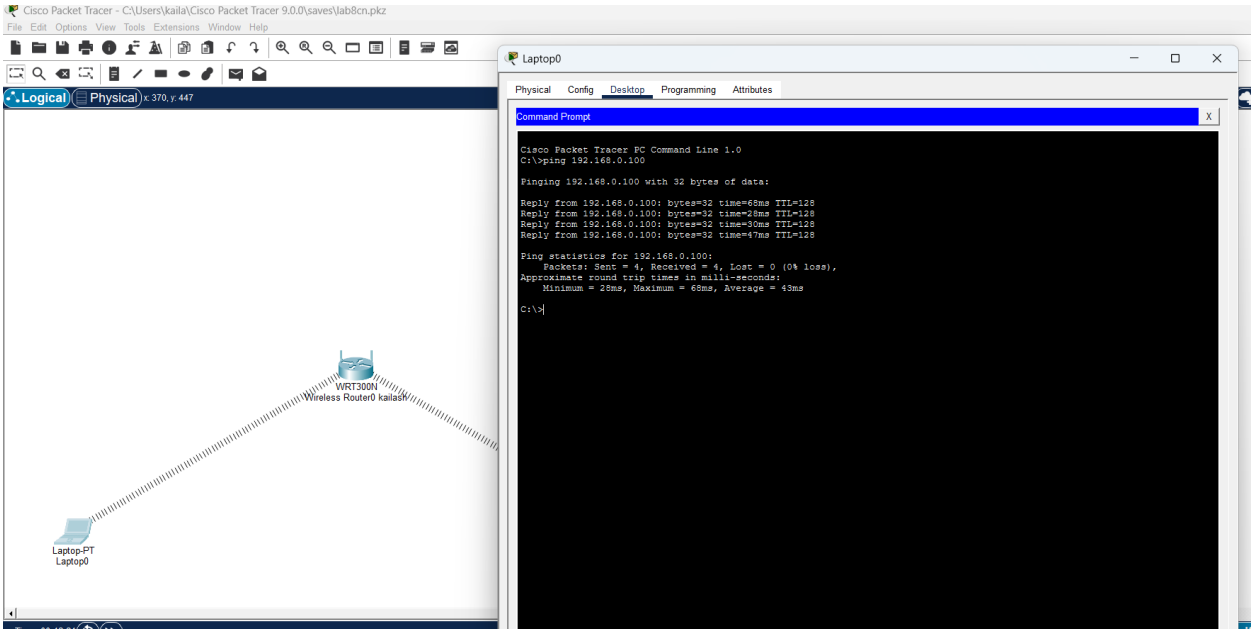
• Encryption: AES • Passphrase: hcoe@123

3. Save the configuration.

Testing

- Testing if both laptops connected to WirelessRouter0

- Ping Test (From Laptop1 to Laptop0)



DISCUSSION

This lab focused on setting up a wireless network and analyzing packet flow in Cisco Packet Tracer. Configuring the WRT300N router allowed multiple devices to connect wirelessly, demonstrating how DHCP simplifies IP management by automatically assigning addresses. Implementing WPA2 encryption highlighted the importance of securing wireless networks, ensuring only authorized devices could access the network and data remained encrypted.

Observing packet flow in simulation mode helped visualize network operations such as DHCP request/acknowledgment exchanges, ARP resolution, and encrypted communication. Troubleshooting minor issues, like incorrect SSID or passphrase, reinforced practical networking skills and understanding of wireless protocols.

RESULTS

All wireless clients successfully connected to the network and obtained unique IP addresses from the DHCP server. WPA2 security was effective, preventing unauthorized access to the network. Ping tests between devices and the router confirmed reliable communication. Simulation mode demonstrated proper packet flow, showing DHCP operations, ARP processes, and encrypted wireless data, verifying that the network was correctly configured and secure.

CONCLUSION

The lab successfully established a functional wireless network with dynamic IP addressing and WPA2 security. Cisco Packet Tracer simulation validated packet-level operations and reinforced the theoretical concepts of wireless networking, DHCP, ARP, and encryption. Students gained practical experience in configuring wireless routers, connecting multiple devices, and monitoring secure network traffic. Overall, the exercise strengthened understanding of wireless network setup, security implementation, and packet analysis.